UPI Fraud Stress Lab

Can an attack campaign reach its target before a detector stops it?

A working, synthetic campaign-level research prototype

prototype/rl-academic-complete | code baseline 92a1296

Offline simulation only. No real payments. Gate C remains HOLD.

Why measure a whole campaign?

A transaction label is not the same as an intervention outcome.

ALLOW	→	FLAG	→	BLOCK
Transfer can proceed.		Alert raised; transfer may proceed.		Simulator stops the campaign.

Measure target completion, first alert, first block and gross simulated debit.

Source: prototype source and retained records

How the prototype works



Feedback: detector → adaptive attacker (public decisions only).

Two saved detectors

XGBoost and Random Forest
Reused artifacts; no retraining required for the demo.

Source: prototype source and retained records

Four workload families

High-value burst · Slow drain
Gradual escalation · Beneficiary hopping

Compare four attacker controls

Static	Family rules that do not respond to detector feedback.
Rule-adaptive	Hand-written rules respond to observed decisions.
Trained RL	Tabular reinforcement learning (RL), trained before evaluation.
Zero-Q control	Same RL action vocabulary without learned Q values.

RL: at most 64 states and 16 amount / pacing / recipient actions.

Source: prototype source and retained records

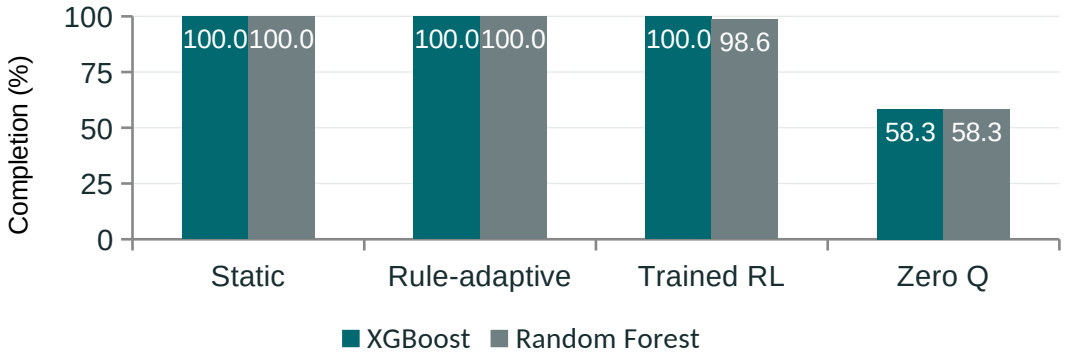
What the retained records show

1,152
attack campaigns

4,189
evaluated transactions

3
evaluation seeds

High-value-burst subset: campaign completion



High-value-burst workload
72 campaigns per arm / detector
3 seed-level replications

Seed SDs: speaker notes.

Historical, confounded comparisons.
Not detector accuracy or proof of RL superiority.

Source: prototype source and retained records

A bounded engineering contribution

Existing methods

Fraud classification
Sequential modelling
Adversarial evasion
Learning-based attackers

Our implemented combination

Campaign + ledger execution
Inspectable tabular policies
Saved checkpoints and raw traces
Reproducible comparisons

Useful research infrastructure, not a first-ever algorithm or superiority claim.

Source: source-grounded literature matrix (review branch)

Run the prototype in a terminal

From a clone of prototype/rl-academic-complete; Linux / WSL2, Python 3.11.

```
python3.11 -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements-milestone1.lock
python -B -s run_prototype.py \
  --detector both --attacker all --seeds 101 \
  --episodes 2 --campaigns 1 --output demo-faculty-01
```

Inspect campaigns.json, transactions.csv, summary.csv and comparison.svg.

Source: prototype source and retained records

What we can defend at review

Implemented	Campaign simulation, two detectors, four attacker controls and raw outputs.
Still limited	Synthetic data; detector-family exclusion unverified; legacy policy comparisons confounded.
Submission pending	The later IEEE draft is AI-assisted. Human authorship review and approved checks remain.

Later dashboard, matched pilot and IEEE paper: review branch

Source: prototype source and retained records